

Multiple Choice

1. **Answer:** A

Options: A) decreased by a factor of 9; B) decreased by a factor of 49; C) decreased by a factor of 81; D) increased by a factor of 9

Note: Correct option: A - decreased by a factor of 9

2. **Answer:** D

Options: A) 0.1 GeV/ c^2 ; B) 0.2 GeV/ c^2 ; C) 0.5 GeV/ c^2 ; D) 1.0 GeV/ c^2

Note: Correct option: D - 1.0 GeV/ c^2

3. **Answer:** D

Options: A) Diode laser; B) Dye laser; C) Free-electron laser; D) Gas laser

Note: Correct option: D - Gas laser

4. **Answer:** B

Options: A) Constant temperature; B) Constant volume; C) Constant pressure; D) Adiabatic

Note: Correct option: B - Constant volume

5. **Answer:** C

Options: A) 1/2; B) 1; C) 3/2; D) 5/2

Note: Correct option: C - 3/2

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Hall coefficient'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Constant volume'.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'The entropy of the system and its environment remains unchanged.'

Long Form Response

11. **Answer:** $0.25 mc^2$. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** Mass of the moon. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key